PharmaLeaf · B.Pharm Semester V · PCI 2026 (NEP 2020)

BPS01T Biomedicinal Chemistry

PharmaLeaf notes for B.Pharm Semester V (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order. For each drug class you get the **classification table, structures of key drugs, mechanism, uses, SAR** (where the syllabus asks) and **synthesis of starred (*) drugs** as reaction schemes.

Syllabus at a glance

BPS01T is a 3-credit core theory course: 3 lectures per week, 45 hours in total, 75 marks (30 sessional + 45 end-semester).

Unit	Title	Topics	Hours
I	Drugs acting on the CNS	General anaesthetics; sedatives and hypnotics (SAR of barbiturates and benzodiazepines); antipsychotics (SAR of phenothiazines); anticonvulsants (SAR)	10
II	Anti-infective chemotherapeutic agents	Antitubercular; antimalarial (SAR of quinolines); antiprotozoal; antiviral; antifungal; anthelmintic; urinary tract anti-infectives (SAR of quinolones)	12
III	Antibiotics and sulfa drugs	Penicillins, cephalosporins, beta-lactamase inhibitors, monobactams, aminoglycosides, tetracyclines, macrolides, chloramphenicol, clindamycin (SAR of penicillins, cephalosporins, tetracyclines); sulfonamides (SAR)	8
IV	Antineoplastic agents	Alkylating agents, antimetabolites, antibiotics, plant products, platinum compounds	6
V	Endocrine drugs, antidiabetics and narcotic analgesics	Sex hormones, ED drugs, contraceptives, corticosteroids, thyroid drugs; insulin and oral antidiabetics; opioids and antagonists (SAR of morphine analogues)	9

Starred (*) drugs whose synthesis is required: ketamine, phenobarbital, diazepam, chlorpromazine, phenytoin, carbamazepine, isoniazid, para-aminosalicylic acid, chloroquine, metronidazole, acyclovir, ketoconazole, albendazole, ciprofloxacin, chloramphenicol, dapsone, mercaptopurine, methotrexate, tolbutamide, fentanyl.

A note on two starred drugs: ketamine and fentanyl are widely misused controlled substances, so PharmaLeaf does not print their synthesis routes. Their structures, SAR and

uses are covered here; learn the synthesis outlines from your prescribed textbook in class.

How to read the structures: carbon atoms are corners and line ends; hydrogens on carbon are not shown. Coloured letters show heteroatoms (**N blue, O red, S yellow-brown, halogens green**).

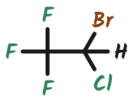
Unit I: Drugs Acting on the Central Nervous System (10 hours)

I.1 General anaesthetics

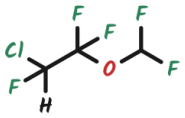
General anaesthetics (GAs): *drugs that produce* ***reversible loss of consciousness and of all sensations****, with* ***analgesia, amnesia, muscle relaxation and loss of reflexes****, so that surgery can be done. Modern* ***balanced anaesthesia*** *combines an i.v. induction agent, an inhalational agent for maintenance, an opioid, a muscle relaxant and premedication.*

Mechanism (current view): *GAs act on specific* ***ion channel proteins****: most* ***enhance GABA-A receptor*** *chloride currents (thiopental, propofol, etomidate, volatile agents, remimazolam); some* ***block NMDA (glutamate) receptors*** *(ketamine, nitrous oxide, xenon); volatile agents also open* ***two-pore K^+ channels*** *and inhibit nicotinic receptors. The older* ***Meyer-Overton theory*** *linked potency to* ***lipid solubility*** *(oil/gas partition coefficient): the more lipid-soluble, the lower the* ***MAC*** *(minimum alveolar concentration) and the more potent.*

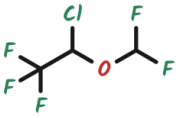
Structures of General Anaesthetics



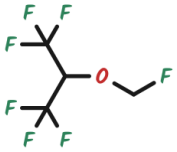
Halothane
halogenated hydrocarbon



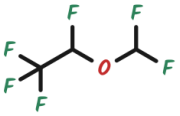
Enflurane
ether



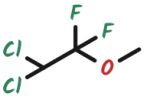
Isoflurane
isomer of enflurane



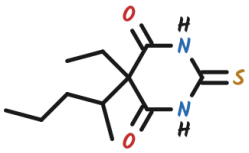
Sevoflurane
fluoromethyl ether



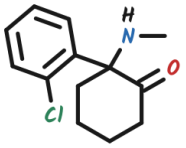
Desflurane
most fluorinated: fastest



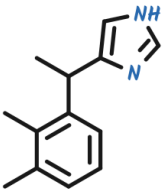
Methoxyflurane
nephrotoxic (F-)



Thiopental
2-thiobarbiturate



Ketamine
arylcyclohexylamine



Dexmedetomidine
alpha-2 agonist (imidazole)

Remember: more fluorine = faster on and off but less potent; 2-thio barbiturate = ultra-short (thiopental)

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Classification

Class	Drugs	Key chemistry
A. Inhalational: halogenated hydrocarbon	**Halothane**	CF_3-CHBrCl (2-bromo-2-chloro-1,1,1-trifluoroethane)